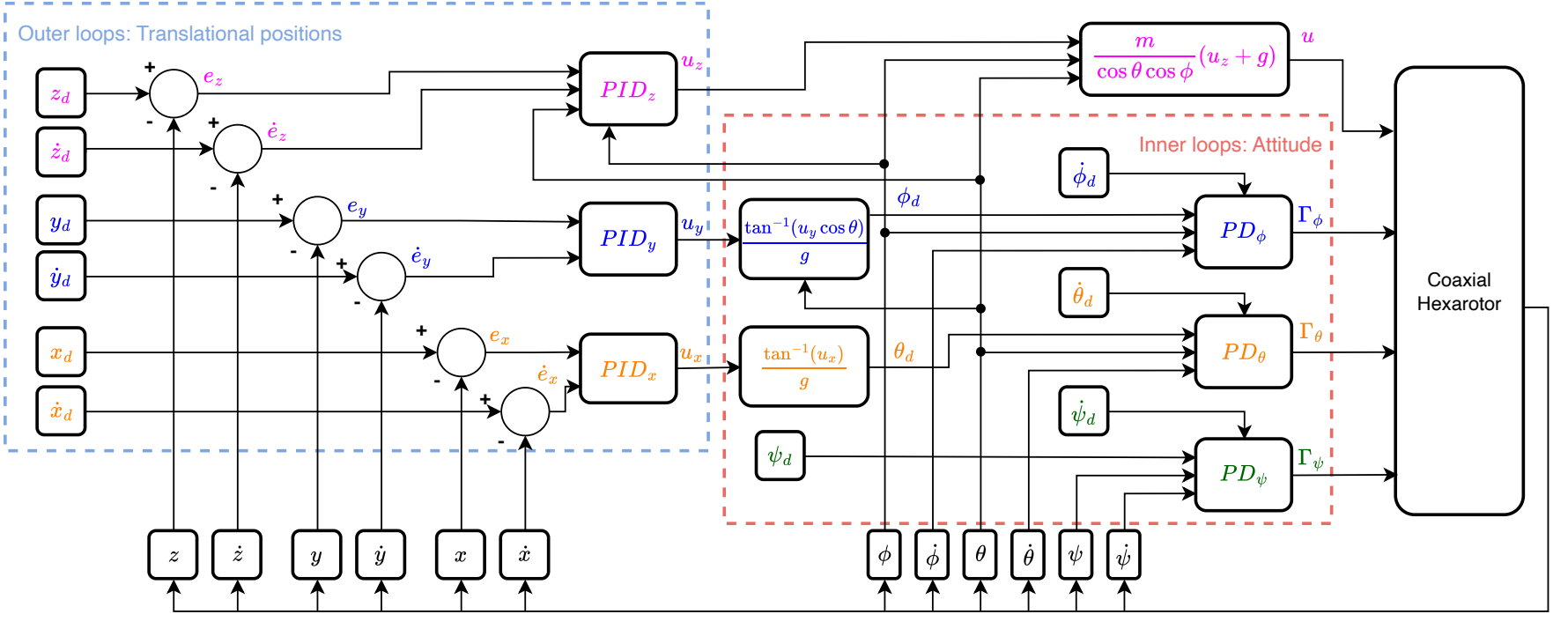

Outer loops: Translational positions
Inner loops: Attitude
z_d
$\dot{z}_d$
y_d
$\dot{y}_d$
x_d
$\dot{x}_d$
e_z
$\dot{e}_z$
e_y
$\dot{e}_y$
e_x
$\dot{e}_x$
PID_z
PID_y
PID_x
u_z
u_y
u_x
$\frac{m}{\cos\theta\cos\phi}(u_z+g)$
u
$\frac{\tan^{-1}(u_y\cos\theta)}{g}$
$\frac{\tan^{-1}(u_x)}{g}$
ϕ_d
θ_d
ψ_d
$\dot{\phi}_d$
$\dot{\theta}_d$
$\dot{\psi}_d$
PD_ϕ
PD_θ
PD_ψ
Γ_ϕ
Γ_θ
Γ_ψ
Coaxial Hexarotor
z
$\dot{z}$
y
$\dot{y}$
x
$\dot{x}$
ϕ
$\dot{\phi}$
θ
$\dot{\theta}$
ψ
$\dot{\psi}$